Exercise Sheet 3

Generative Models and Naive Bayes

Probabilistic Machine Learning

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3.A A Posterior and Its Peak

Exercise 3.A.1: Two Analysts, One Campaign

Same data: 60 of 400 converted. Analyst A uses a flat prior $\mathrm{Beta}(1, 1)$; analyst B, from earlier campaigns, uses $\mathrm{Beta}(20, 80)$.

1. Write down both posteriors and their means.
2. How many "pseudo-observations" is B's prior worth? Why do the two analysts nearly agree?
3. The data arrived in two batches: 25 of the first 200, 35 of the second 200. Show that updating twice gives the same posterior as updating once.
4. What is A's probability that the next user converts?

Exercise 3.A.2: Peak, Mean and a Prior's Peak

Returning users, $\theta = P(\text{user returns})$, prior $\text{Beta}(a, b)$, h of n users return.

1. Maximise $\log p(D \mid \theta) + \log p(\theta)$ and show $\hat{\theta}_{\text{MAP}} = \frac{h+a-1}{n+a+b-2}$.
2. **0 of 4** users return. Give the MLE, the MAP under $\text{Beta}(2, 2)$ and the posterior mean under $\text{Beta}(2, 2)$. Which one would you report to a product manager, and why?
3. The posterior mean under a flat prior is $\frac{h+1}{n+2}$. Under which prior is the **MAP** equal to it?
4. After 3 of 3 with a flat prior the posterior is $\text{Beta}(4, 1)$, with density $4\theta^3$. Compute its central 95% interval.

3.B The Generative Route

Exercise 3.B.1: Counting Parameters

A fraud classifier (two classes) uses four attributes: **payment** (card, invoice, wallet), **country** (domestic, EU, other), **new device** (yes/no) and **night** (yes/no).

1. How many free numbers does the full generative model need — the class prior and one joint $p(x \mid y)$ per class?
2. How many does naive Bayes need?
3. For p yes/no attributes and K classes, give both counts as formulas.
4. From which p on does the full joint need more than a million numbers *per class*?

Exercise 3.B.2: Fitting Is Counting

An attribute takes k values. Within one class we observed it n times, value v exactly n_v times ($\sum_v n_v = n$). Model it as categorical with parameters $\theta_1, \ldots, \theta_k \geq 0, \sum_v \theta_v = 1$.

1. Write down the log-likelihood $\ell(\theta)$ of the n i.i.d. observations.
2. For $k = 2$, substitute $\theta_2 = 1 - \theta_1$ and show that the MLE is $\hat{\theta}_1 = n_1/n$ — the h/n of Lecture 03.
3. For general k, maximise ℓ under the constraint $\sum_v \theta_v = 1$ (Lagrange multiplier) and show $\hat{\theta}_v = n_v/n$.
4. Why can naive Bayes fit every attribute's table separately, while a full joint $p(x \mid y)$ could not be fitted attribute by attribute?

3.C Naive Bayes by Hand

Exercise 3.C.1: Twelve Transactions

Flags: new device D, foreign IP F, night N, high amount H (1 = yes).

#	D	F	N	H	fraud?	#	D	F	N	H	fraud?
1	1	1	1	1	yes	7	1	0	0	0	no
2	1	1	0	1	yes	8	0	0	0	0	no
3	1	0	1	0	yes	9	0	1	0	0	no
4	0	1	1	1	yes	10	0	0	0	0	no
5	0	0	0	0	no	11	1	0	1	0	no
6	0	0	1	0	no	12	0	0	0	0	no

1. Estimate $P(\text{fraud})$ and $P(\text{flag} = 1 \mid \text{class})$ for all four flags by counting.
2. Compute $P(\text{fraud} \mid x)$ for $x_1 = (D, F, N, H) = (1, 0, 0, 0)$.
3. Do the same for $x_2 = (0, 0, 0, 1)$. What goes wrong, and why?
4. Repeat 2 and 3 with Laplace smoothing on the flags (keep the prior unsmoothed). Which result changes a lot, which only a little?

Exercise 3.C.2: Smoothing Is a Prior

A yes/no flag was seen h times among n frauds.

1. With a $\mathrm{Beta}(1,1)$ prior on $\theta = P(\mathrm{flag} \mid \mathrm{fraud})$, what is the posterior (Lecture 03)? Show that its **mean** is Laplace's $(h+1)/(n+2)$.
2. What is the posterior's **peak** for $h = 0$? What does that say about "keep the peak"?
3. "Add-α" smoothing uses $(h+\alpha)/(n+2\alpha)$. Which prior belongs to it? Compute the estimate for $h = 0, n = 100$ and $\alpha = 0.5, 1, 10$.
4. A merchant category with $k = 10$ values; one category never occurred in 100 frauds. What does Laplace smoothing give, and which prior is behind it?